

Thomas Landzaat

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B.S. ECE (4.0 GPA) — Seeking Summer 2026 FPGA Internships

Skills

Hardware Design: SystemVerilog, Verilog, NoC, AXI-4, FPGA

Embedded: C, C++, ESP32, Arduino

Tools & Software: Vivado, Quartus, QuestaSim, Cadence Xcelium, Genus, Innovus, SimVision

Lab Skills: Signal Processing, Oscilloscopes, Circuit Design, Soldering, PCB Design

Projects

FPGA Engineer Intern, Tektronix – Beaverton, OR

June 2025 – Aug 2025

- Designed a parametrized memory Built-In Self-Test engine (BIST) in SystemVerilog supporting multiple memory depths and AXI 4 configurations
- Achieved 500 MHz timing closure through multi-stage pipelining and verified correctness with QuestaSim & on-hardware ILA
- Collaborated on NoC architecture enabling 120 GB/s sustained BIST write throughput by optimizing NMU placement for minimal routing congestion

Projects

SCARA Robotic Drawing Arm

Jan 2025 – March 2025

- Designed, programmed, and built a Wi-Fi-enabled robotic arm controlled via a custom Python GUI, capable of interpreting G-code and graphical drawings in real-time
- Developed firmware in C++ to calculate inverse kinematics for precise arm movements
- Engineered and ordered a professional-grade PCB using KiCad; modeled and 3D-printed robotic arms and a custom enclosure to achieve seamless integration between mechanical and electronics.

Four-bit Serial I/O Multiplier

February 2025 – March 2025

- Designed a 4×4 unsigned serial-in/serial-out multiplier in Verilog and synthesized the design in Cadence Genus using 0.13 μm CMOS flow
- Implemented a FSM Controller to manage serial loading, multiplication and timed shift out with ready and done flags
- Completed place and route in Cadence Innovus and performed post layout STA to verify timing

ESP32 Temperature-Controlled Smart Aquarium Heater

Oct 2024 – Nov 2024

- Flashed smart plug running Esp8266 with custom firmware (Tasmota)
- Configured the ESP32 with a temperature sensor and established mesh networking for seamless device communication such as toggling the smart plug based on the aquarium temperature
- Eliminated the need for an external MQTT broker by utilizing direct HTTP commands within the mesh network also increasing the reliability of the system when unexpected power loss occurs

Education

Oregon State University

4.0 GPA

B.S. Electrical and Computer Engineering, Minor in Computer Science

Expected March 2026

M.Eng. Electrical and Computer Engineering

Expected March 2027

Favorite Classes: VLSI Design Automation, High Performance Computer Architecture, Energy-Efficient VLSI Design